CITS4012 Project
TweetQA

Group 5

1 Introduction

Question answering over tweets requires understanding short, informal texts containing names, hashtags, mentions, spelling variation and author metadata. The system must identify which information answers a particular question while distinguishing it from other plausible details in the same tweet. Our aim was to design and train an attention-based neural model from scratch, investigate its components through controlled experiments and compare it with complementary baselines. We developed an extractive model based on bidirectional attention flow (BiDAF; Seo et al., 2017), adapted with character representations, tweet-specific features and gated self-attention. We then investigated probability ensembling and out-of-fold candidate reranking. The selected system achieved validation ROUGE-L of 47.11 and final test ROUGE-L of 56.17. It substantially outperformed lexical rules and a simplified neural baseline, although pretrained BERTweet, T5-small and Qwen performed better.

2 Methodology

2.1 Task Formulation and Silver Span Supervision

Given a tweet (T) and question (Q), our main model predicts a continuous answer span within the original tweet. Extractive prediction preserves names and wording and avoids generating unsupported text. However, it cannot freely paraphrase or combine information from separated locations. TweetQA provides reference answers without span annotations. We therefore construct silver training labels by enumerating candidate spans of at most 15 tokens and comparing them with the reference using normalised token F1. Ties are resolved using ROUGE-L, shorter span length and earlier position. Content-token rules restrict eligible endpoints. An example contributes to the extractive training loss when its best alignment reaches F1 of at least 0.5. This retains 8,057 examples from the training partition. The selected model uses the single best span as its target. Earlier experiments also investigated soft supervision distributed across multiple overlapping spans. Silver alignment enables supervised span learning, but introduces noise and excludes poorly aligned examples. These restrictions are particularly important for paraphrased and non-extractive reference answers.

2.2 Token Representations

Each token is represented by concatenating:

- A 100-dimensional word embedding.
- A 150-dimensional character-CNN representation.
- Eight tweet-specific token features.

The resulting representation has 258 dimensions. Character embeddings have dimension 32, with convolution widths of 3, 4 and 5 and 50 filters per width. ReLU and max pooling produce the character representation. Word lookup uses normalised tokens, while the character CNN processes lowercased raw tokens. This combination shares word representations across common formatting variations while retaining information about individual names and spellings. The eight features indicate mentions, hashtags, URLs, emoji, numeric tokens, initial capitalisation, byline membership and non-stopword overlap with the other sequence. Byline information is retained because tweet authors can be the correct answer to questions. Word embeddings are initialised using FastText trained on the permitted training partition (Bojanowski et al., 2017). No externally pretrained embeddings or neural checkpoint initialise the main model.

2.3 Contextual Encoding

Two highway layers process the token representations. A shared bidirectional LSTM then encodes the question and tweet separately, using 128

hidden units in each direction and producing 256-dimensional contextual vectors.

2.4 Bidirectional Attention Flow

Following BiDAF, let $\mathbf{h}_i$ denote the contextual representation of tweet token i, and $\mathbf{u}_j$ denote question token j. Their similarity is

$$S_{ij} = \mathbf{w}^\top [\mathbf{h}_i; \mathbf{u}_j; \mathbf{h}_i \odot \mathbf{u}_j], \tag{1}$$

where concatenation is denoted by semicolons and elementwise multiplication by $\odot$.

Context-to-question attention produces a question summary for each tweet token:

$$\mathbf{c}_i = \sum_j \mathrm{softmax}_j(S_{ij})\mathbf{u}_j. \tag{2}$$

Question-to-context attention produces a shared tweet summary:

$$\mathbf{q} = \sum_i \mathrm{softmax}_i \left(\max_j S_{ij} \right) \mathbf{h}_i. \tag{3}$$

The combined representation is

$$\mathbf{G}_i = [\mathbf{h}_i; \mathbf{c}_i; \mathbf{h}_i \odot \mathbf{c}_i; \mathbf{h}_i \odot \mathbf{q}]. \tag{4}$$

It is projected from 1,024 to 256 dimensions. Padding masks exclude invalid positions.

Context-to-question attention conditions each tweet position on the question, while question-to-context attention identifies tweet information relevant to the question as a whole.

2.5 Gated Self-Attention and Answer Prediction

Four-head self-attention operates over the question-aware tweet representations. It enables interactions between separated positions, including the tweet body and author byline.

Inspired by gated self-matching approaches (Wang et al., 2017), a learned gate controls the attention update:

$$\mathbf{g} = \sigma\left(\mathbf{W}[\mathbf{x}; \mathbf{a}] + \mathbf{b}\right), \tag{5}$$

$$\mathbf{y} = \mathrm{LayerNorm}\left(\mathbf{x} + \mathrm{Dropout}(\mathbf{g} \odot \mathbf{a})\right), \tag{6}$$

where $\mathbf{a}$ is the self-attention output.

A two-layer modelling BiLSTM feeds the start-position head, and a further BiLSTM feeds the end-position head. Training minimises the sum of start and end cross-entropies:

$$\mathcal{L} = -\log p_s(i^*) - \log p_e(j^*). \tag{7}$$

At inference, decoding selects

$$(\hat{i}, \hat{j}) = \underset{\substack{0 \le i \le j < T \\ j - i < 15}}{\arg\max} \left[\log p_s(i) + \log p_e(j)\right]. \tag{8}$$

Stored character offsets map the selected span back to unmodified tweet text.

2.6 Probability Ensembling and OOF Reranking

The selected system combines three independently trained v2 checkpoints. Their start probabilities and end probabilities are averaged separately. The ten strongest legal spans under these averaged distributions become reranker candidates.

A LambdaMART reranker (Burges, 2010) uses 30 features describing neural confidence, candidate rank, span length and location, lexical overlap, token and boundary types, question type and body/byline region. Prediction-time features use no reference answers.

Reranker training candidates are generated using three tweet-grouped outer folds of the training partition. Each proposing model trains on the other folds, with an inner grouped split for early stopping, and predicts its held-out fold. Consequently, each reranker-training question receives candidates from a model that did not train on its tweet. Training references supply graded candidate relevance labels.

These OOF fold models are separate from the three final QA checkpoints. The selected reranker transfers from single-model OOF candidates to final ensemble candidates; this combination was explicitly evaluated on validation.

3 Experimental Setup

3.1 Dataset and Partitioning

The supplied training file contains 10,692 questions over 5,119 tweets. Removing 46 exact duplicate rows leaves 10,646 questions.

We construct a deterministic tweet-grouped split using seed 4012, approximately stratified by question type. Table 1 summarises the resulting partitions.

Partition	Questions	Tweets
Training	9,570	4,608
Validation	1,076	511
Final test	1,086	583

Table 1: Dataset partitions. Tweet groups are disjoint between training and validation.

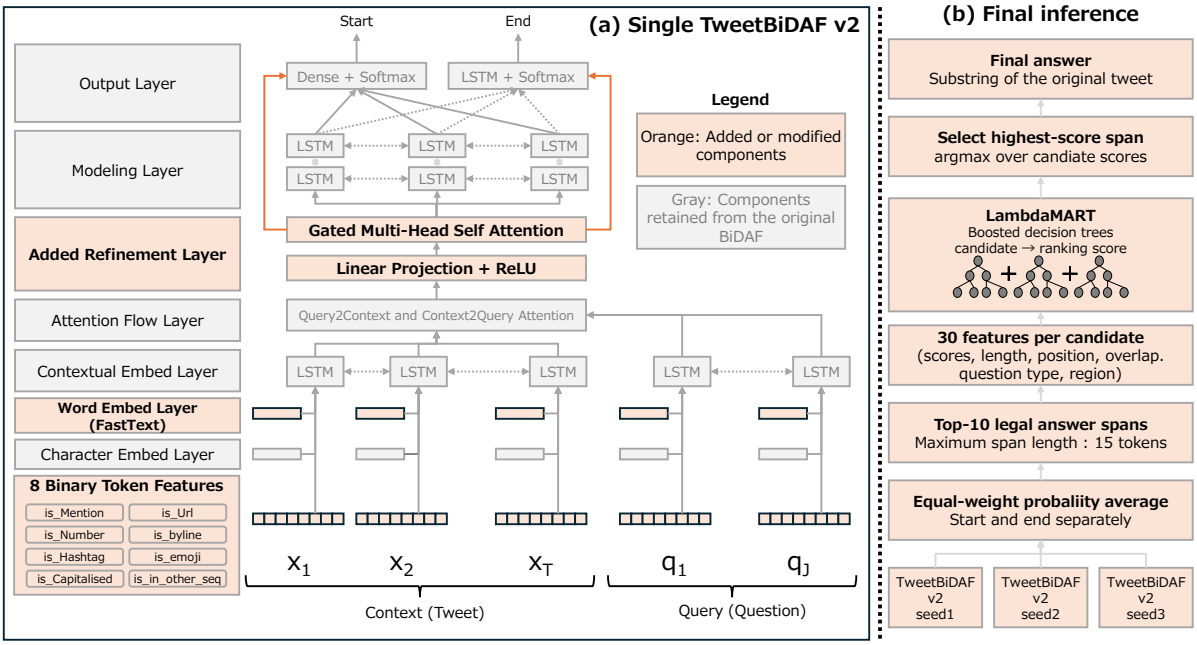


Figure 1: Architecture of TweetBiDAF v2 and the final inference pipeline. (a) Orange highlights components added or modified relative to BiDAF; gray indicates retained components. (b) Three independently trained seeds are combined by averaging start and end probabilities separately. The top ten legal spans are ranked by an OOF-trained LambdaMART reranker using 30 features, and the highest-scoring span is extracted from the original tweet.

Grouping prevents questions about the same tweet appearing in both training and validation. The supplied training and test files share three tweet texts, so complete train–test context separation cannot be claimed.

Validation has one reference per question, while test has two. We preserve original row indices because question IDs are not necessarily unique.

3.2 Training and Hyperparameters

The main model uses Adam with learning rate 5×10^{-4}, batch size 32, dropout 0.2 and gradient clipping at 5. Training allows 30 epochs, with early stopping after four epochs without improvement in validation ROUGE-L.

A one-factor development sweep investigated learning rate, hidden size, dropout, maximum span length and in-domain FastText initialisation. This was not an exhaustive factorial search. Three seeds were used for the principal component experiments to measure training variability.

LambdaMART uses 150 trees, learning rate 0.05, at most 15 leaves, maximum depth four, minimum child size 50 and L2 regularisation of 1.0.

3.3 Baseline Models

B1: Lexical rules. Short windows are ranked using proximity to question words and answer-type cues. Training-only selection fixes maximum length five, zero length penalty and type bonus one. This tests the value of learning beyond lexical heuristics.

B2: Simplified neural model. This baseline removes character features, normalisation/flags, self-attention and question-to-context attention. It retains a reduced recurrent extractive model and tests the combined value of the main adaptations. Its three seeds are evaluated separately.

B3: BERTweet. We fine-tune `vinai/bertweet-base` with a span head, testing domain-relevant pretrained representations (Nguyen et al., 2020). B3 uses seed one, three epochs, learning rate 3×10^{-5}, batch size 16 and maximum input length 128. Subword span predictions map back to original tweet offsets.

B4: T5-small. We fine-tune a text-to-text model to generate answers (Raffel et al., 2020). This tests an output format capable of producing text absent from the tweet. B4 uses seed one, five epochs, learning rate 3×10^{-4}, batch size 16, source length

160 and target length 24. Inference uses four beams and at most 16 new tokens.

B5: Qwen. We use `Qwen2.5-1.5B-Instruct` (Qwen Team, 2024) with four fixed demonstrations and no fine-tuning. This tests instruction-following and in-context learning. A fixed instruction requests a short answer, followed by greedy generation of at most 16 new tokens and fixed output cleanup. The prompt and checkpoint revision are recorded in the notebook.

All numerical baseline results come from the conducted experiments. Differences in pretraining, scale and output format mean these comparisons contextualise performance rather than isolate individual architectural effects.

3.4 Evaluation Metrics

We report BLEU-1, METEOR, ROUGE-L ($\beta = 1.2$), normalised exact match and token F1. These measure unigram precision, lexical alignment, sequence overlap, complete agreement and token overlap respectively. Each metric takes the best reference score per question, then averages across questions and multiplies by 100. BLEU-1 uses NLTK method-1 smoothing; METEOR uses NLTK and WordNet.

Each metric takes its best value over available references for each question, then averages across questions and multiplies by 100. BLEU-1 uses NLTK sentence BLEU with method-1 smoothing; METEOR uses NLTK and WordNet.

Partial-credit metrics help distinguish boundary mistakes from unrelated answers, while exact match measures complete normalised agreement. However, lexical metrics can undervalue valid paraphrases or reward incomplete answers.

4 Results and Analysis

4.1 Overall Performance Comparison

Table 2 presents results calculated from predictions on test data.

Our system improves ROUGE-L over B1 by 30.03 points and over mean B2 by 21.03. These differences include representation design, attention, ensembling and reranking.

All pretrained baselines perform better. Qwen exceeds our system by 19.54 ROUGE-L points, while BERTweet and T5-small reach 71.17 and 70.79. These results indicate the limits of the small supervised system relative to pretrained models, but do not separately identify the effects of scale, prior data, domain adaptation or generation.

The test ROUGE-L of 56.17 exceeds validation ROUGE-L of 47.11 despite unchanged checkpoints. The examples differ, and two test references offer more opportunities for a match than the single validation reference. The difference is notan improvement between evaluations.

4.2 Controlled Ablation Studies

Table 3 uses the original matched three-seed validation batch, separate from the later final ensemble checkpoints.

Removing normalisation and flags causes the largest reduction. This supports the combined importance of tweet-specific preprocessing and structure, although the experiment does not isolate normalisation from individual flags.

Removing the character CNN tests whether raw spelling information contributes beyond word embeddings. Its smaller decrease supports retaining character representations. Removing self-attention tests interactions between tweet positions, while removing question-to-context attention tests the global question-conditioned tweet summary. Both comparisons favour the full design.

Wide soft supervision underperforms single-span targets. Distributing probability across partially overlapping spans may weaken boundary specificity. Narrower soft-target experiments also failed to surpass single-span training. This is a plausible interpretation of the observed results, rather than direct proof of boundary blurring.

Relevant conditions were kept matched within the batch, including the data split and training protocol. Nevertheless, component removal changes parameter counts and optimisation, and three seeds provide only a limited variability estimate.

4.3 Ensembling and Reranking

Candidate analysis showed that useful answers often appeared below the first neural prediction. For one current-gated checkpoint, a reference-assisted Top-10 oracle achieved ROUGE-L 67.16, compared with 44.14 for Top-1. This oracle uses gold answers and is a diagnostic ceiling, not deployable performance.

The selected reranker improves ensemble ROUGE-L by 1.39 points. A paired bootstrap resampling whole tweets gives a 95% interval of approximately $[+0.27, +2.52]$. Resampling tweets

System	BLEU-1	METEOR	ROUGE-L	EM	F1
B1: lexical rules	21.76	17.16	26.14	14.46	26.16
B2: simplified neural	30.73 ± 0.98	27.81 ± 0.49	35.14 ± 1.01	21.82 ± 0.96	34.96 ± 1.06
Our v2 ensemble + OOF reranker	50.99	42.81	56.17	39.96	56.13
B3: fine-tuned BERTweet	67.55	54.40	71.17	57.83	71.11
B4: fine-tuned T5-small	67.11	54.65	70.79	53.04	70.71
B5: four-shot Qwen	**72.08**	**56.35**	**75.72**	**63.54**	**75.93**

Table 2: Final test performance on 1,086 questions. B2 reports mean $\pm$ sample standard deviation across three individual seeds. Our selected system is one ensemble plus reranker. Bold marks the strongest result.

Variant	ROUGE-L	Δ
Full single-span v2	44.51 ± 0.11	—
No character CNN	43.56 ± 0.39	-0.94
No normalisation/flags	32.82 ± 0.69	-11.68
No self-attention	42.52 ± 0.84	-1.99
No Q2C attention	43.50 ± 0.70	-1.01
Wide soft span targets	42.95 ± 0.47	-1.56
B2: combined simplification	30.14 ± 0.29	-14.37

Table 3: Controlled validation ablations: mean $\pm$ sample standard deviation across three seeds. Changes are variant minus full model, calculated before rounding.

Validation configuration	ROUGE-L
Three-model ensemble, Top-1	45.72
Ensemble + original OOF reranker	**47.11**
Extra boundary/confidence features	46.70
Matched ensemble-OOF training	46.50

Table 4: Validation comparison of ensemble and reranking configurations.

preserves dependence among questions about the same context.

Matched ensemble-OOF training differs from the selected system by -0.61 points, with an interval of $[-1.46, +0.22]$. This does not establish a clear difference, but provides no convincing reason to replace the selected configuration. These intervals do not correct for repeated validation-based selection.

4.4 Qualitative Analysis of Attention

The attention visualisations examine the first selected v2 seed. They are not attention maps for the ensemble or reranker.

Successful example. **Question:** "who was was asked how oklahomans should feel about kevin durant?"

Tweet: "Sam Presti was asked how Oklahomans should feel about Kevin Durant. Pretty remarkable answer: Royce Young (@royceyoung) July 5, 2016"

Reference: "sam presti"

Prediction: "Sam Presti"

The boundary distributions favour the correct two-token answer despite another person appearing in the byline. Question-to-context attention also assigns weight to other question-related words, showing that correct prediction does not require every attention component to concentrate exclusively on the answer.

Failed example. **Question:** "who can not confirm this?"

Tweet: "DUDES. I can't confirm this. But I also can't deny this... Lauren Graham (@thelaurengraham) October 19, 2015"

Reference: "lauren graham"

Prediction: "DUDES"

The context-to-question map is broadly similar across tweet positions, while the start/end heads favour the beginning of the tweet. Relevant author information is present but not selected. This illustrates body/byline confusion and insufficient discrimination between a salient opening word and the person responsible for the statement.

These patterns reveal associations, but cannot establish that an individual attention weight caused a prediction. Controlled retraining ablations provide stronger evidence about component usefulness.

Fresh validation analysis of the selected ensemble identifies 752 EM failures. Automatic rules assign 461 to references absent as literal tweet substrings, 143 to partial token overlap and 148 to no token overlap. These are descriptive categories, not manually verified causes.

4.5 Reproducibility and Limitations

The final notebook loads frozen artifacts, generates predictions and calculates all five metrics without training during its normal run. It contains implementation code, dependencies, training records and evaluation outputs. The selected system reproduces

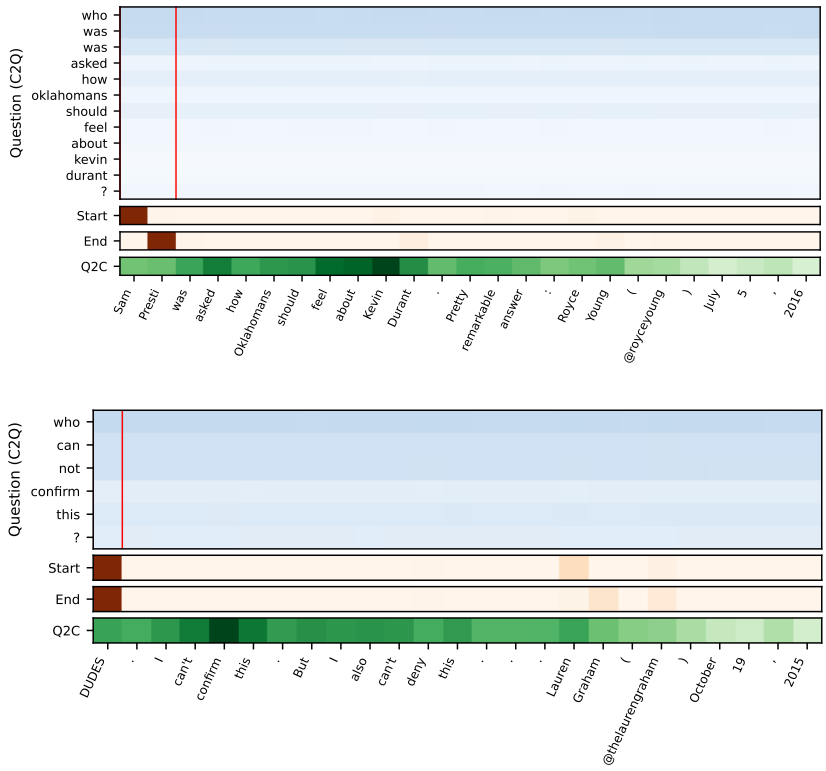


Figure 4: Successful and failed single-seed validation examples. Include context-to-question attention and start/end distributions. The corresponding questions, tweets, references and predictions are provided in the text. These visualisations describe the base model rather than the final ensemble/reranker.

all 1,076 original validation answer strings and metrics before test inference.

B3 and B4 use predetermined three- and five-epoch runs with seed one. These fitting runs exclude test data. Their complete fine-tuned weights and tokenizers are retained, and both reproduce all 1,076 saved validation answers after reloading. Their validation ROUGE-L scores are 59.98 and 57.30.

Limitations include noisy silver alignment, exclusion of poorly aligned training examples, restricted extractive output and transfer from single-model OOF candidates to ensemble candidates. Repeated development on one validation set may overstate selected gains. Supplied train–test context overlap and automatic error categories also limit generalisation claims.

5 Conclusion

We developed and evaluated an attention-based TweetQA system combining character information, tweet-specific features, recurrent encoding, bidirectional attention and gated self-attention. Controlled experiments support these adaptations, with normalisation and flags producing the largest measured contribution.

Single-span supervision outperformed the tested soft-target alternatives. Probability ensembling and OOF reranking improved the selected validation configuration, which achieved final test ROUGE-L of 56.17. This substantially exceeds lexical and simplified neural baselines, while pretrained systems remain stronger, led by Qwen at 75.72.

The project demonstrates meaningful learning and measurable design contributions while exposing persistent challenges in answer expressiveness, entity selection and evaluation discipline. Future work should investigate nested grouped development, blinded error annotation and controlled approaches to non-extractive prediction.

Team Contributions

James Hudibyu: Implemented the initial data-processing and evaluation pipeline, main neural architecture and baseline workflows. Conducted the original training, hyperparameter investigations and component ablations, and developed the single-span v2 model.
Kaz Morita: Extended the experimental investigation through attention analysis, alternative model designs, candidate-oracle studies and out-of-fold span reranking. Evaluated ensembling and reranker variants.
Prabhjot Kaur: Reproduced the selected system's validation results and conducted final test evaluations for the main model and baselines. Retrained BERTweet and T5-small using the original fixed settings and verified that reloading the saved models reproduced their validation predictions. Integrated and tested the submission notebook, organised supporting artifacts, and wrote the final report.

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